

BENJAMIN MARIANO

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Education

University of Texas at Austin

PhD, Department of Computer Science with advisor Işıl Dillig

Sep. 2019 – Present

Austin, Texas

University of Maryland, College Park

M.Sc., Department of Computer Science with advisor Jeff Foster

Sep. 2017 – May 2019

College Park, Maryland

University of Maryland, College Park

B.Sc., Department of Computer and Electrical Engineering

Sep. 2013 – May 2017

College Park, Maryland

Research Experience

University of Texas at Austin

Research Assistant with Işıl Dillig

Sep. 2019 – Dec. 2024

Austin, Texas

- Designed system for automatically deobfuscating software using program synthesis
- Created neural-guided program synthesizer for automatically translating functional code to imperative code
- Performed large-scale study of common loop semantics summaries in smart contracts
- Helped design type system for detecting overflow vulnerabilities in smart contracts
- Worked on tool for automatically translating Apache Spark RDD programs to SQL
- Helped create SmartLTL, a tool for automatically verifying liveness properties in smart contracts

Max Planck Institute

Research Intern with Maria Christakis

May. 2019 – Sep. 2019

Kaiserslautern, Germany

- Designed optimization methodology for balancing precision and scalability in abstract interpretation
- Performed empirical study on real-world programs to measure precision/performance tradeoffs

University of Maryland, College Park

Research Assistant with Jeff Foster

Sep. 2017 – May 2019

College Park, Maryland

- Developed and tested new models of the Java Standard Library for synthesis optimization
- Experimented with new axiomatic programming paradigms for program synthesis

Industry Experience

Veridise

Vice President of Research and Development

May 2022 – Present

Austin, Texas

- Leading fuzzing team aimed at developing automated testing software for DeFi applications
- Acting as project lead on multiple audits of client smart contract and blockchain code
- Overseeing design and development of Chainsaw, a coverage-guided fuzzer for blockchain protocols which has found over a dozen critical bugs in production-level software to date.
- Serving as project lead for OrCa, an oracle-guided fuzzer for testing smart contracts against user-provided logical specifications.

Intel

Research Engineer with Justin Gottschlich

Oct. 2021 – May 2022

Austin, Texas

- Worked in the Machine Programming team developing automated techniques for learning syntax-driven code translators
- Designed technique using enumerative program synthesis for discovering succinct code translation rules

Prime Solutions

Software Engineering Intern

Summer 2015, Summer 2016

Columbia, Maryland

- Designed, developed, and tested simulation of malicious protocols in C and Python
- Automated analysis of local WiFi traffic to determine device-specific information

Teaching Experience

University of Texas at Austin <i>Teaching Assistant for Automated Logic and Reasoning with Işil Dillig</i>	Jan. 2022 – May 2022 <i>Austin, Texas</i>
University of Texas at Austin <i>Teaching Assistant for Discrete Math with Işil Dillig</i>	Sep. 2019 – Dec. 2019 <i>Austin, Texas</i>
University of Maryland, College Park <i>Co-teacher for Functional Pearls with Cameron Moy</i>	Jan. 2018 – May 2018 <i>College Park, Maryland</i>
University of Maryland, College Park <i>Teaching Assistant for Advanced Functional Programming with Niki Vazou</i>	Sep. 2017 – Dec. 2017 <i>College Park, Maryland</i>

Awards and Service

PLDI 2022 Artifact Evaluation Committee	2022
POPL 2019 PLMW Travel Scholarship	2019
University of Maryland Honor’s College Member	2013-2017
University of Maryland Dean’s Scholarship	2013-2017
University of Maryland Dean’s List	2013-2017

Publications

Control-Flow Deobfuscation using Trace-Informed Compositional Program Synthesis. <u>Benjamin Mariano</u> , Ziteng Wang, Shankara Pailoor, Christian Collberg, Işil Dillig	OOPSLA 2024
Automated Translation of Functional Big Data Queries to SQL. Guoqiang Zhang, <u>Benjamin Mariano</u> , Xipeng Shen, Işil Dillig	OOPSLA 2023
Automated Transpilation of Imperative to Functional Code Using Neural-Guided Program Synthesis. <u>Benjamin Mariano</u> , Yanju Chen, Yu Feng, Shuvendu K. Lahiri, Işil Dillig	OOPSLA 2022
SolType: Refinement Types for Arithmetic Overflow in Solidity. Bryan Tan, <u>Benjamin Mariano</u> , Shuvendu K. Lahiri, Işil Dillig, Yu Feng	POPL 2022
Automatically Tailoring Abstract Interpretation to Custom Usage Scenarios. Muhammad Numair Mansur, <u>Benjamin Mariano</u> , Maria Christakis, Jorge A. Navas, Valentin Wüstholtz	CAV 2021
SmartPulse: Automated Checking of Temporal Properties in Smart Contracts. Jon Stephens, Kostas Ferles, <u>Benjamin Mariano</u> , Shuvendu K. Lahiri, Işil Dillig	S&P 2021
Demystifying Loops in Smart Contracts. <u>Benjamin Mariano</u> , Yanju Chen, Yu Feng, Shuvendu K. Lahiri, Işil Dillig	ASE 2020
Program Synthesis with Algebraic Library Specifications. <u>Benjamin Mariano</u> , Josh Reese, Siyuan Xu, ThanhVu Nguyen, Xiaokang Qiu, Jeffrey S. Foster, Armando Solar-Lezama	OOPSLA 2019